

Ex. No.:10

Roll No.:22bcs119

Date :

System Testing

Aim

To write test cases and perform system testing for Library Management system by manual testing.

Test cases

Scenario-1 Search Book Use Case

Test ID	User Interface / Method	Input Given	Expected Output	Actual Output	Testing Status
TC_S01	Search Book Interface	Book Title: "The Great Gatsby"	List of matching books with details (Title, Author, ISBN)	List of matching books with details	Pass
TC_S02	Search Book Interface	Book Title: "Nonexistent Book"	Message: "No books found"	Message: "No books found"	Pass
TC_S03	Search Book Interface	Invalid ISBN: "123"	Error: "Invalid ISBN format"	Error: "Invalid ISBN format"	Pass
TC_S04	Search Book Interface	Author: "J.K. Rowling"	List of books by J.K. Rowling with details	Incorrect book details	Fail

Scenario-2 Check Availability Use Case

Test ID	User Interface / Method	Input Given	Expected Output	Actual Output	Testing Status
TC_A01	Check Availability Interface	Book ID: "12345"	Status: "Available" or "Checked Out" with due date	Status: "Available"	Pass
TC_A02	Check Availability Interface	Book ID: "99999" (non-existent)	Error: "Book not found"	Error: "Book not found"	Pass
TC_A03	Check Availability Interface	Book ID: ""	Error: "Book ID cannot be empty"	Error: "Book ID cannot be empty"	Pass
TC_A04	Check Availability Interface	Book ID: "12345" (all copies checked out)	Status: "No copies available"	Incorrect availability status	Fail

Scenario-3 Borrow Book Use Case

Test ID	User Interface / Method	Input Given	Expected Output	Actual Output	Testing Status
TC_B01	Borrow Book Interface	User ID: "U001", Book ID: "12345"	Confirmation: "Book borrowed, due in 14 days"	Confirmation: "Book borrowed, due in 14 days"	Pass
TC_B02	Borrow Book Interface	User ID: "U001", Book ID: "12345" (already borrowed)	Error: "Book is not available"	Error: "Book is not available"	Pass
TC_B03	Borrow Book Interface	User ID: "U002", Book ID: "67890" (borrow limit reached)	Error: "Borrow limit reached"	Error: "Borrow limit reached"	Pass
TC_B04	Borrow Book Interface	User ID: "U003", Book ID: "54321" (suspended user)	Error: "User account suspended"	Book borrowed successfully	Fail

Scenario-4 Return Book Use Case

Test ID	User Interface / Method	Input Given	Expected Output	Actual Output	Testing Status
TC_R01	Return Book Interface	Book ID: "12345", Return Date: Today	Confirmation: "Book returned, no fines"	Confirmation: "Book returned, no fines"	Pass
TC_R02	Return Book Interface	Book ID: "12345" (already returned)	Error: "Book already returned"	Error: "Book already returned"	Pass
TC_R03	Return Book Interface	Book ID: "99999" (non-existent)	Error: "Book not found"	Error: "Book not found"	Pass
TC_R04	Return Book Interface	Book ID: "12345" (not borrowed)	Error: "Book not borrowed"	Incorrect fine applied	Fail

Scenario-5 Check Borrow Duration Use Case

Test ID	User Interface / Method	Input Given	Expected Output	Actual Output	Testing Status
TC_D01	Check Borrow Duration Interface	User ID: "U001", Book ID: "12345"	Duration: "Borrowed on [date], Due on [date]"	Duration: "Borrowed on [date], Due on [date]"	Pass
TC_D02	Check Borrow Duration Interface	User ID: "U001", Book ID: "99999" (not borrowed)	Error: "No borrow record found"	Error: "No borrow record found"	Pass
TC_D03	Check Borrow Duration Interface	User ID: "", Book ID: "12345"	Error: "User ID cannot be empty"	Error: "User ID cannot be empty"	Pass
TC_D04	Check Borrow Duration Interface	User ID: "U002", Book ID: "12345" (overdue)	Duration: "Borrowed on [date], Overdue by [days]"	Incorrect duration displayed	Fail

Result

Thus the test cases were written and system testing for Library Management system is done by manual testing